

# Selective Offloading

Scalar SMU + RVV proposed design

new

reused

— solid=data

--- dashed=software/MMIO sequencing

## Proposed Design Boundary

TCDM offsets, N, D, mode=1, start

### MMIO Register / Command Interface

TCDM offsets, N, D, mode=1 (scalar-only)

status: busy / done / error

### Software/Core

MMIO write + start  
wait busy → done  
load weights  
issue RVV update

busy / done / error

NEW

### Scalar SMU

#### State-dependent Scalar Recurrence

$(m\_old, l\_old, m\_tile, l\_tile)$

$\max + 2 \times \text{EXP LUTs}$

$l\_new \rightarrow \text{Reciprocal LUT} \rightarrow w\_old, w\_tile$

after done: software loads weights from TCDM; issues RVV update

$m/l + \text{weights}$

REUSED

### Reuse Existing RVV Datapath Regular $O[D]$ Vector Update

$O\_new[j] = w\_old \times O\_old[j] + w\_tile \times O\_tile[j]$

VLSU load → vector multiply/FMA → store

$O$  vectors

### Shared TCDM (software-visible state)

$m/l + w\_old, w\_tile + O$  vectors